

Single-channel EEG seizure detection on CHB-MIT: accuracy, model size and efficiency

Bold = best in column. Underlined = second best. Higher is better for ACC and SEN; lower is better for parameters, stored footprint and MACs. Every row is CHB-MIT. Footprints marked *est.* are derived from a parameter count rather than reported by the paper, and are excluded from the ranking. EpiSepNet-5K is this group's own earlier model, kept in the table as the design it succeeds. Weight memory and total on-chip memory are different quantities: most papers report only the first. **DFP** is dynamic fixed point, a power-of-two scale: its footprint EQUALS plain integer at the same width, because the scale is per tensor rather than per value. What it changes is the datapath — requantising between layers becomes a shift instead of a multiply.

Method	Dataset	Model	Protocol / test exposure	ACC (%)	SEN (%)	Parameters	Precision	Weight memory	Other efficiency
WearSeizure-1D (this work)	CHB-MIT (13 single-channel cases)	Separable 1D-CNN, k5 + dilation 1/2/4/8/16; 1 channel; 4-s windows	Leakage-safe : split by recording before filtering; thresholds frozen on val; 185.0 h continuous test; 66 folds x 3 seeds	98.88	94.95 event level; 60.33 segment level	11,786	DFP8 chosen on measured loss: 0.00 pp of event sensitivity against FP32 on all 66 folds, through the real integer datapath	10.87 KiB weights (DFP8) + 1.76 KiB biases (32-bit)	<u>585,920 MACs (thop) / 489,600 conv+fc; FAR 0.25/h</u> total on-chip 18.62 KiB = weights + biases + 6.00 KiB <u>peak activation</u> <u>3.6% of XC7Z020 BRAM</u>
WearSeizure-1D , same code under the segment-split protocol	CHB-MIT (13 single-channel cases)	As above, retrained under the leaky protocol	<i>Segment-level random split</i> ; normalised on all data; threshold on test — i.e. the protocol common to the rows below. 99.6% of test windows have a near-duplicate in training	99.68	92.29 segment level	11,786	FP32 (measured)	as above	Same network; only the split rule differs from the row above
EpiSepNet-5K (INT16) earlier model, same group	CHB-MIT	BatchNorm-folded separable 1D-CNN	—	90.04	90.76	4,900	INT16	<u>10.0 KB package; ~9.8 KB raw INT16</u>	99.9743% agreement with FP32; 2.81x smaller package
EpiSepNet-5K (FP32) earlier model, same group	CHB-MIT	Separable 1D-CNN; 17-channel raw EEG; 2-s windows	—	90.07	90.76	<u>5,010</u>	FP32	28.1 KB checkpoint; ~20.0 KB raw FP32	Reference model
Chung et al. 2024 [1]	CHB-MIT (13 cases)	Stacked 2D-CNN, parallel 1x3 and 1x5 kernels; 1 clinician-selected channel	Segment-level 70/20/10 split per patient; ~91 h	98.18	99.62 event level; 96.76 segment level	116,700	FP32	454 KB at FP32 rebuilt from the paper; 116 226 params vs 116 700 reported, +0.4%	FAR 0.22/h; detection delay 3.3 s; specificity 98.19% weights + activations: 510 KB at FP32 = 91% of XC7Z020 BRAM (rebuilt from the paper, peak 14 464 values)
Werner et al. (TC-ResNet4) [2]	CHB-MIT	16-channel TC-ResNet4 with fixed-point inference	—	95.28	92.34	9,840	4-bit	~4.92 KB weight-only (4-bit)	337,968 MACs; 495 nW average power
Zhu et al. 2021 [3]	CHB-MIT (23 channels)	7-layer 1D-CNN, 5x1 conv, shift-register PE array	80/10/10 split; normal class down-sampled 200:1	97.35	94.32	7,010	fixed-point	<i>est.</i> 28 KB at FP32, 7 KB at 8-bit	6.32 MOPs; 170 us/inference @ 200 MHz; FPGA Xilinx Zynq ZC706
Li et al. 2022 [4]	CHB-MIT + Bonn + SWEC-ETHZ	1D-CNN with parallel convolutional layers	80/20 + 5-fold CV; GAN-synthesised preictal segments	<u>99.01</u>	<u>99.24</u>	10,778	analog RRAM	<i>est.</i> 43 KB at FP32	1.13 us parallelised; 7.21 W; ASIC 22 nm FDSOI + RRAM crossbar
Ferrara et al. [5]	CHB-MIT	Patient-specific two-channel lightweight CNN	—	99.00	67.00	9,500	—	51 KB model	Balanced accuracy 83.0%; 0.10 FP/h
REST-RS [6]	CHB-MIT	Graph-based residual state-update model	—	—	—	9,300	—	0.037 MB, ~37 KB	AUROC up to 93.5%; 1.314 ms inference
SlimSeiz [7]	CHB-MIT	Eight-channel convolution and Mamba network (prediction task)	—	94.80	95.50	21,200	—	<i>est.</i> 84.8 KB at FP32	Specificity 94.0%
RGF-Model [8]	CHB-MIT	Multi-teacher knowledge-distillation model	—	98.92	98.54	82,000	FP32	0.33 MB (330 KB)	Specificity 99.11%; AUC 98.96%
Wang et al. (MSCA) [9]	CHB-MIT	Inverted residual CNN with multi-scale channel attention	—	98.70	98.30	88,000	—	<i>est.</i> 352 KB at FP32	2.68M MACs; specificity 99.10%
Ahlawat (INT8) [10]	CHB-MIT	Quantized common-channel 1D-CNN with operator fusion	—	—	—	2014	INT8	0.44 MB (440 KB) at INT8; 1.63 MB at FP32	Up to 2.8x speedup; up to 64% estimated energy reduction

Reading the table. Protocols differ between rows, and most papers do not state theirs in enough detail to compare. This work is the only row evaluated with splits taken by recording, thresholds frozen on a held-out validation partition, and 185 h of continuous test exposure. Under the segment-level random split used by much of this literature, the same code and data reach 92.29 % segment sensitivity instead of 60.33 % — a 31-point difference caused by the protocol alone, because 99.6 % of test windows then have a near-duplicate in training. Accuracy is nearly blind to this: at 0.62 % ictal prevalence a model that never predicts a seizure already scores 99.38 %.

Table 2 — what the single-channel constraint costs

Three arms of the **same** model, identical except for how many channels it reads, so the difference is attributable to the channel count and nothing else. 66 folds x 3 seeds = 198 runs per arm; montages taken verbatim from Chung et al. so the comparison lines up with their published ablation channel-for-channel. **All three train from scratch per fold — no cohort pre-training, no distillation — which is why the 1-channel arm reads 91.79 % here and 94.95 % in Table 1. Compare the arms with each other, never with Table 1.**

Channels	SEN event (%)	SEN segment (%)	ACC (%)	AUROC	FAR/h	Parameters	MACs	Total on-chip	Runs on the accelerator?
1 (deployed)	91.79	49.36	98.96	0.8870	<u>0.2798</u>	11,578	585,920	18.62 KiB	yes - 1,024 of 16,384 cells
4	<u>93.31</u>	<u>59.36</u>	<u>99.10</u>	<u>0.9237</u>	0.3334	<u>11,746</u>	<u>671,936</u>	<u>20.79 KiB</u>	yes - 4,096 of 16,384 cells
18	95.20	70.51	99.37	0.9586	0.2625	12,530	1,073,344	35.55 KiB	no - needs 18,432 cells

Paired bootstrap against the 18-channel arm, clustered by patient. 1 channel: **−3.41 pp** event sensitivity, 95 % CI [−5.83, −0.93] — excluding zero, so the cost is established; and −21.16 pp at segment level, CI [−26.28, −16.47]. 4 channels: −1.89 pp event, CI [−5.89, +1.97] — *spans zero*, so four channels are not measurably worse than eighteen. **Post-processing absorbs 84 % of the deficit**: 21.16 points at segment level become 3.41 at event level, because smoothing, hysteresis and run-length filtering integrate the score across consecutive windows. That is why a single-electrode device is viable at all. Chung et al. report the 18-to-1 penalty as 1.9 points of segment sensitivity; measured without their segment-level random split it is **21.2**. Accuracy moves **0.41 pp** across an eighteen-fold change in input channels while segment sensitivity moves 21.2 — a third independent demonstration that it cannot be used to compare detectors on this data. AUROC, being threshold-free, is not blind.

Table 3 — activation memory, layer by layer

Definition: **the largest sum of the feature maps of two adjacent layers**, plus any skip connection that must stay live across them. A layer reads its input while writing its output, so both are resident at the same instant. **This network has no skip connections** — DepthwiseSeparableConv1d is depthwise → pointwise → BN → ReLU and the multi-scale block in k5-only mode is a single branch, so neither adds its input back — and there is therefore nothing to add. Bytes at DFP8, one byte per activation.

Layer	Input map	Output map	Input + output
stem.0	<u>1,024 B</u>	4,096 B	5,120 B
b1.dw	4,096 B	2,048 B	6,144 B
b1.pw	2,048 B	4,096 B	6,144 B
b2.dw	4,096 B	2,048 B	6,144 B
b2.pw	2,048 B	3,072 B	5,120 B
b3.dw	3,072 B	1,536 B	4,608 B
b3.pw	1,536 B	2,048 B	3,584 B
b4.dw	2,048 B	1,024 B	3,072 B
b4.pw	1,024 B	1,536 B	2,560 B
context.0.dw	1,536 B	1,536 B	3,072 B
context.0.pw	1,536 B	2,048 B	3,584 B
context.1.dw	2,048 B	2,048 B	4,096 B
context.1.pw	2,048 B	2,048 B	4,096 B
gap	2,048 B	<u>64 B</u>	<u>2,112 B</u>
fc	64 B	2 B	66 B

Peak = 6,144 B = 6.00 KiB, at b1.dw. Total on-chip = 10.87 KiB weights + 1.76 KiB biases + 6.00 KiB activations = **18.62 KiB**. Two details that would otherwise be asked about. **Biases are stored at 32 bit**, not at the data width: the PE loads a bias as the accumulator's initial value, so it lives at the accumulator's fixed point, and counting them at 8 bit would understate the total by 1.3 KiB. And **the accelerator allocates 8.00 KiB rather than 6.00**, because it uses a ping-pong pair of banks — layers alternate between them instead of being packed — so each bank must hold the largest map it ever sees. 6.00 KiB is what the network requires; 8.00 KiB is what is on the die, for a total of 20.62 KiB. The 18-channel arm needs 22.00 KiB of activation and does not fit the 16,384-cell feature-map memory at all.

References

[1] Y. G. Chung, A. Cho, H. Kim, K. J. Kim. Single-channel seizure detection with clinical confirmation of seizure locations using CHB-MIT dataset. Front. Neurol. 15:1389731, 2024.
[2] J. Werner, B. Kohli, P. Palomero Bernardo, C. Gerum et al. TC-ResNet for EEG seizure detection.
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[4] C. Li, C. Lammie, X. Dong, A. Amirsoleimani, M. Rahimi Azghadi, R. Genov. Seizure Detection and Prediction by Parallel Memristive Convolutional Neural Networks. IEEE TBioCAS, 2022.
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[9] Wang et al. Lightweight Seizure Detection Based on Multi-Scale Channel Attention, 2023.
[10] K. Ahlawat. Efficient EEG Seizure Detection Using INT8 Quantization, Channel Pruning, and Spiking Neural Networks. arXiv:2607.16296, 2026.